# Worksheet 2.4 — Linear and Quadratic Functions

Mathematics: Analysis & Approaches SL (IB) · Unit 2: Functions

## The Big Ideas

Vertex form reveals a parabola's turning point immediately.

- Line through two points: $m = \dfrac{y_2 - y_1}{x_2 - x_1}$, then $y - y_1 = m(x - x_1)$.
- Vertex form: $y = a(x - h)^2 + k$, vertex $(h, k)$.
- Complete the square to convert $y = ax^2 + bx + c$ into vertex form.

## Worked Examples

### Example 1 — Equation of a line

Find the equation of the line through $(-1, 3)$ and $(4, 18)$.

***Solution.*** $m = \dfrac{18 - 3}{4 - (-1)} = \dfrac{15}{5} = 3$. $y - 3 = 3(x + 1) \Rightarrow y = 3x + 6$.

**Conclusion:** $y = 3x + 6$.

### Example 2 — Complete the square

Convert $y = x^2 - 10x + 18$ to vertex form.

***Solution.*** Half of $-10$ squared is 25: $y = (x^2 - 10x + 25) - 25 + 18 = (x - 5)^2 - 7$.

**Conclusion:** $y = (x - 5)^2 - 7$, vertex $(5, -7)$.

### Example 3 — Vertex form when $aeq1$

Convert $y = 2x^2 + 8x + 3$ to vertex form.

***Solution.*** Factor 2 from the first two terms: $y = 2(x^2+4x)+3 = 2(x^2+4x+4-4)+3 = 2(x+2)^2-8+3$.

**Conclusion:** $y = 2(x + 2)^2 - 5$.

### Example 4 — Build a quadratic from vertex and a point

Find the equation of a parabola with vertex $(-2, 5)$ through $(1, -13)$.

***Solution.*** $y = a(x + 2)^2 + 5$. Substitute $(1, -13)$: $-13 = a(9) + 5 \Rightarrow a = -2$.

**Conclusion:** $y = -2(x + 2)^2 + 5$.

### Example 5 — Line through vertex and a root

A parabola has vertex $(3, -4)$ and an $x$-intercept at $x = 5$. Find its equation.

***Solution.*** $y = a(x - 3)^2 - 4$. Substitute $(5, 0)$: $0 = a(4) - 4 \Rightarrow a = 1$.

**Conclusion:** $y = (x - 3)^2 - 4$.

### Example 6 — Finding where a line meets a parabola

Find the intersection point(s) of $y = x + 1$ and $y = x^2 - 2x - 3$.

***Solution.*** Set equal: $x + 1 = x^2 - 2x - 3 \Rightarrow x^2 - 3x - 4 = 0 \Rightarrow (x - 4)(x + 1) = 0 \Rightarrow x = 4$ or $x = -1$.

**Conclusion:** intersections at $(4, 5)$ and $(-1, 0)$.

### Example 7 — Perpendicular line through a parabola's vertex

Find the equation of the line perpendicular to $y = 2x + 1$ passing through the vertex of $y = (x - 3)^2 + 5$.

***Solution.*** Vertex is $(3, 5)$. Perpendicular slope: $-\frac{1}{2}$. Line: $y - 5 = -\frac{1}{2}(x - 3)$.

**Conclusion:** $y = -\frac{1}{2}x + 6.5$.

### Example 8 — Application — maximizing revenue

A shop's weekly revenue is $R(p) = -4p^2 + 120p$, where $p$ is the price in dollars. Convert to vertex form and find the price that maximizes revenue.

***Solution.*** $R(p) = -4(p^2 - 30p) = -4(p^2 - 30p + 225 - 225) = -4(p - 15)^2 + 900$.

**Conclusion:** vertex $(15, 900)$: maximum revenue \$900 at price \$15.

### Example 9 — Finding a quadratic from three points (via a system)

Find the equation $y = ax^2 + bx + c$ of the parabola through $(0, 1)$, $(1, 4)$, $(2, 11)$.

***Solution.*** **Step 1:** $(0, 1)$ gives $c = 1$.

**Step 2:** $(1, 4)$: $a + b + 1 = 4 \Rightarrow a + b = 3$. $(2, 11)$: $4a + 2b + 1 = 11 \Rightarrow 4a + 2b = 10 \Rightarrow 2a + b = 5$.

**Step 3:** Subtract: $a = 2$, so $b = 1$.

**Conclusion:** $y = 2x^2 + x + 1$.

## Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

### Question 1

Find the equation of the line through $(2, -1)$ and $(6, 11)$.

### Question 2

Convert $y = x^2 + 6x + 2$ to vertex form.

### Question 3

Convert $y = 3x^2 - 12x + 7$ to vertex form.

### Question 4

State the vertex and direction of opening of $y = -\frac{1}{2}(x - 4)^2 + 9$.

### Question 5

Find the equation of a parabola with vertex $(1, -4)$ through $(3, 4)$.

## Question 6

A ball's height is $h(t) = -4.9(t - 1.5)^2 + 11.5$; find its maximum height and when it occurs.

## Question 7 — Challenge

Convert $y = -2x^2 + 4x + 1$ to vertex form.

## Question 8

Find the intersection point(s) of $y = 2x - 3$ and $y = x^2 - 4$.

## Question 9

A parabola has vertex $(2, 3)$ and an $x$-intercept at $x = 5$. Find its equation.

## Question 10

A company's profit is $P(x) = -3x^2 + 90x$ for $x$ units sold (in hundreds); convert to vertex form and find the maximizing quantity and max profit.

## Question 11 — Challenge

Find the equation $y = ax^2 + bx + c$ through $(0, 2)$, $(1, 7)$, $(2, 16)$.

## Question 12 — Challenge

A parabola has $x$-intercepts at $x = 2$ and $x = 8$ and passes through $(5, -9)$. Find its equation in factored form, then convert to vertex form.

## Question 13 — Challenge

Find the equation of the line perpendicular to $y = -3x+2$ passing through the vertex of $y = (x+1)^2-6$.

# Question Recap & Answer Key

## Question Recap

1. Find the equation of the line through $(2, -1)$ and $(6, 11)$.
2. Convert $y = x^2 + 6x + 2$ to vertex form.
3. Convert $y = 3x^2 - 12x + 7$ to vertex form.
4. State the vertex and direction of opening of $y = -\frac{1}{2}(x-4)^2 + 9$.
5. Find the equation of a parabola with vertex $(1, -4)$ through $(3, 4)$.
6. A ball's height is $h(t) = -4.9(t-1.5)^2 + 11.5$; find its maximum height and when it occurs.
7. Convert $y = -2x^2 + 4x + 1$ to vertex form.
8. Find the intersection point(s) of $y = 2x - 3$ and $y = x^2 - 4$.
9. A parabola has vertex $(2, 3)$ and an $x$-intercept at $x = 5$. Find its equation.
10. A company's profit is $P(x) = -3x^2 + 90x$ for $x$ units sold (in hundreds); convert to vertex form and find the maximizing quantity and max profit.
11. Find the equation $y = ax^2 + bx + c$ through $(0, 2)$, $(1, 7)$, $(2, 16)$.
12. A parabola has $x$-intercepts at $x = 2$ and $x = 8$ and passes through $(5, -9)$. Find its equation in factored form, then convert to vertex form.
13. Find the equation of the line perpendicular to $y = -3x+2$ passing through the vertex of $y = (x+1)^2 - 6$.

## ✓ Answer Key

1. $m = 3$, $y = 3x - 7$
2. $y = (x+3)^2 - 7$
3. $y = 3(x-2)^2 - 5$
4. vertex $(4, 9)$, opens downward
5. $4 = a(4) - 4 \Rightarrow a = 2$: $y = 2(x-1)^2 - 4$
6. max height 11.5 m at $t = 1.5$ s
7. $y = -2(x^2 - 2x) + 1 = -2(x-1)^2 + 2 + 1 = -2(x-1)^2 + 3$
8. $2x - 3 = x^2 - 4 \Rightarrow x^2 - 2x - 1 = 0 \Rightarrow x = 1 \pm \sqrt{2}$; points $(1+\sqrt{2}, 2\sqrt{2}-1)$ and $(1-\sqrt{2}, -2\sqrt{2}-1)$
9. $0 = a(3)^2 + 3 \Rightarrow a = -\frac{1}{3}$: $y = -\frac{1}{3}(x-2)^2 + 3$
10. $P(x) = -3(x-15)^2 + 675$: max profit \$675 (hundred-dollar units) at $x = 15$
11. $c = 2$; $a+b = 5$, $4a+2b = 14 \Rightarrow 2a+b = 7$; subtract: $a = 2, b = 3$: $y = 2x^2 + 3x + 2$
12. $y = a(x-2)(x-8)$; at $x = 5$: $-9 = a(3)(-3) = -9a \Rightarrow a = 1$: $y = (x-2)(x-8) = x^2 - 10x + 16 = (x-5)^2 - 9$
13. vertex $(-1, -6)$; perpendicular slope $\frac{1}{3}$; $y + 6 = \frac{1}{3}(x+1) \Rightarrow y = \frac{1}{3}x - \frac{17}{3}$